



...Hidden image analysis site...

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Abstract:

In an era where digital communication and image sharing have become ubiquitous, the inadvertent exposure of sensitive information embedded in images poses a significant threat to personal privacy and security. This project aims to develop an innovative application designed to analyze images and detect sensitive information, such as phone numbers, addresses, hidden IDs, bank account details, or other private data, before they are shared. By leveraging advanced technologies like Optical Character Recognition (OCR), image processing, and pattern recognition, the application ensures that users are alerted to potential privacy risks and provided with tools to sanitize or remove such information. The application operates through a multi-step process: first, it extracts text from images using OCR tools like Tesseract; second, it identifies sensitive data by applying pattern-matching algorithms, such as regular expressions (Regex), to detect specific formats like phone numbers, email addresses, or bank account numbers; and third, it offers users the ability to blur, pixelate, or completely remove the detected sensitive information using image processing libraries like OpenCV and Pillow. This proactive approach empowers users to take control of their digital privacy and prevents accidental data leaks.

The project addresses a critical need in today's digital landscape, where the consequences of sharing sensitive information can range from identity theft to financial fraud. By integrating cutting-edge technologies into a user-friendly interface, the application not only enhances privacy and security but also raises awareness about the importance of scrutinizing digital content before sharing it. This tool is designed to be scalable, adaptable, and accessible, making it suitable for both individual users and organizations seeking to protect sensitive data. Ultimately, this project represents a significant step forward in safeguarding digital privacy and ensuring secure communication in an increasingly connected world.

Table of content:

Comments by Supervisor(s) / Examiner(s).....	2
Abstract.....	3
Chapter 1 : Introduction.....	6
1.1 Problem Statement.....	7
1.2 Problem Solution.....	7
1.3 Project Scope.....	8
1.4 Development Tools	8
1.5 Aims & Objectives	9
1.6 Purpose of the project	10
1.7 Research Thesis.....	10
1.8 Summary	11
Chapter 2:Background.....	11
2.1 Introduction.....	11
2.2 History and Evolution of Programs.....	11
2.3 Key Objectives.....	12
2.4 Summary.....	12
Chapter 3:Literature Review.....	13
3.1 Introduction.....	13
3.2 Existing Websites.....	13
3.2.1 StegExpose – Best for Detecting Steganography.....	13
3.2.2 FotoForensics – Best for Image Forensic Analysis.....	13
3.2.3 Forensically – Best for Digital Image Investigation.....	14
3.2.4 DeepAI Image Recognition – Best for AI-Powered Analysis.....	14
3.2.5 ExifTool.....	14
3.2.6 StegOnline.....	14
3.2.7 Image Edited.....	14
3.2.8 Google Vision API.....	14

3.2.9 Tineye.....	15
3.2.10 IZITRU.....	15
3.2.11 Amazon Rekognition.....	15
3.2.12 Google lens.....	15
3.2.13 Microsoft Azure Computer Vision.....	15
3.2.14 Tesseract OCR.....	16
3.3 conclusion.....	16
Chapter 4: requirement Analysis.....	16
4.1 introduction	16
4.2 Requirement Specifications	16
4.2.1 Functional Requirment	16
4.2.2 Non-functional Requirements	17
4.3 Entity Relationship Diagram (ERD).....	18
4.3.1 General ERD.....	19
4.4 Conclusion.....	19

Chapter 1: Introduction

In today's digital age, sharing images has become an integral part of communication, whether through social media, messaging apps, or professional platforms. However, many users are unaware that images often contain hidden sensitive information, such as phone numbers, addresses, ID card details, or even bank account numbers, embedded within them. Accidentally sharing such information can lead to privacy breaches, identity theft, or financial fraud. To address this growing concern, this project proposes the development of an intelligent application designed to analyze images for sensitive data before they are shared. The tool leverages cutting-edge technologies, including Optical Character Recognition (OCR), image processing, and pattern recognition, to detect and sanitize sensitive information. By providing users with a warning and the option to blur or remove such data, the application ensures that privacy is maintained and risks are minimized. This project is particularly relevant in a world where digital privacy is increasingly under threat. With the rise of data breaches and cyberattacks, individuals and organizations need tools to proactively protect their information. By combining powerful libraries like OpenCV, Tesseract OCR, Pillow, and Regex, this application offers a robust and user-friendly solution to a critical problem.

The goal of this project is not only to enhance privacy and security but also to raise awareness about the importance of scrutinizing digital content before sharing it. By integrating advanced technologies into a simple and accessible tool, this application empowers users to take control of their digital footprint and safeguard their sensitive information effectively.

1.1 Problem Statement

As the use of social media and online platforms increases, users are increasingly sharing images containing sensitive personal information, intentionally or unintentionally.

These images may contain phone numbers, addresses, ID card details, bank account numbers, etc. that could be misused if made public.

The lack of tools to automatically identify and protect such sensitive data before it is published poses a significant privacy and security risk.

The problem lies in the inability of users to manually detect and remove sensitive information from images, which can lead to inadvertent data disclosure, identity theft, and invasion of privacy. Despite the availability of optical character recognition (OCR) technology, there is no effective solution to automatically analyze and process images to detect and render invisible sensitive data such as phone numbers, addresses, and bank account information before it is shared.

Our solution is to develop an application that automatically analyzes images to detect sensitive data such as phone numbers, addresses, ID numbers and bank account details and alerts the user before the image is shared.

The solution provides an important level of protection for users in an increasingly digital world, preventing accidental data leaks.

1.2 Problem Solution

The proposed solution is an intelligent application that analyzes images for sensitive information, detects it using advanced technologies, and provides users with options to sanitize or remove the data before sharing. Knowing that their private information is protected, this addresses a critical gap in digital privacy by providing a tool that proactively detects and sanitizes sensitive information in images. By combining OCR, pattern recognition, and image processing technologies, the application ensures that users can share images safely and securely, reducing the risk of privacy breaches and data misuse. This project not only solves a practical problem but also promotes awareness about the importance of protecting personal information in the digital world.

1.3 project Scope

This project focuses on developing a smart application designed to protect user privacy when sharing images online. The application will utilize AI and OCR (Optical Character Recognition) technologies to analyze images and identify any sensitive information within them. The app will first extract text from images using Tesseract OCR, then apply Regex and AI-based detection methods to recognize patterns of sensitive data, such as phone numbers, addresses, bank account details, and identification card numbers

The application will allow users to upload images from their devices or take photos directly with the camera. Once an image is uploaded, the app will extract the text and analyze it for sensitive data by comparing the detected text with predefined patterns. This includes detecting local and international phone numbers, addresses (using keyword matching), bank account numbers, and other personally identifiable information like ID card numbers or email addresses

Upon identifying any sensitive data, the application will notify the user with the details of the detected information. The user will then have the option to either blur or remove the sensitive data automatically using image processing tools like OpenCV and Pillow. The final image, with sensitive information concealed, can then be saved and shared securely

1.4 Development Tools

Tesseract OCR:

Utilized for extracting visible text from images, enabling subsequent analysis of potential sensitive information.

Regular Expressions (Regex):

Applied to identify and extract specific data formats—such as phone numbers and email addresses—from the text.

OpenCV:

Used for image pre-processing tasks, including quality enhancement and region of interest isolation, to facilitate more effective text extraction.

StegExpose:

Utilized to perform statistical analysis on image data, detecting anomalies that may indicate the presence of hidden (steganographic) information.

ExifTool:

Used to examine and extract metadata from images, thereby uncovering any concealed or altered information that is not visible in the image content.

1.5 Aims and objective

sensitive information, including phone numbers and addresses, and provides warnings to users prior to sharing such images. This will enable individuals to share images confidently without compromising their personal information. Furthermore, the website will serve to educate users on the significance of safeguarding sensitive information in digital content

Objectives/

Image Analysis

Employ Optical Character Recognition (OCR) techniques to detect text within images.

Identify sensitive information

such as :

Phone numbers

Addresses

Identity card numbers

Bank account numbers

Other private information

User-Friendly Interface

Design an intuitive and accessible interface that allows users to upload and analyze images effectively

Provide clear warnings and actionable suggestions for addressing detected sensitive information

Real-Time Processing

Implement real-time image analysis to deliver immediate feedback to users

Optimize the website to ensure rapid and efficient processing of image

1.6 Purpose of the project

1. Privacy Protection:

- Prevent unintended leakage of sensitive information in images, such as addresses, phone numbers, or any other personal data that may be hidden or invisible.

2. Detect Tampering and Forgery:

- Analyze images to detect modifications (such as Photoshop) and determine whether images have been manipulated for illegal or misleading purposes.

3. Enhance Cybersecurity:

- Detect Steganography that may be hidden within images and used to transmit confidential or malicious information without the user's knowledge.

1.7 Research Thesis

This thesis consists of eight chapters, one introduction, chapter two is background, chapter three is a literature review, chapter four is system analysis, chapter five is system design, chapter six is a system implementation, chapter seven is testing and chapter eight is conclusion

1.8 Summary

This chapter introduced the importance of developing an application to detect and protect sensitive information in images before sharing. It highlighted privacy risks such as identity theft and data breaches due to unintentional disclosure. The proposed solution leverages OCR, image processing, and pattern recognition to detect and sanitize personal data. Additionally, the chapter outlined the project's scope, explaining how the application will analyze images, alert users, and provide options to blur or remove sensitive details, ensuring safer digital sharing.

Chapter 2: Background

2.1 Introduction

This project is dedicated to developing an innovative and detailed image analysis system specifically designed to detect sensitive data and hidden messages within various images. The system will leverage cutting-edge technologies such as Optical Character Recognition (OCR) and advanced steganalysis techniques to fulfill its objectives. By seamlessly integrating these technologies, the project aspires to provide a comprehensive solution that not only analyzes but also actively safeguards user data, thereby enhancing privacy and security in digital communications. The ability to uncover hidden information and protect sensitive data is increasingly essential in our data-driven world, and this initiative seeks to fill that critical need.

2.2 History and Evolution of Programs

Optical Character Recognition (OCR)

The evolution of Optical Character Recognition (OCR) technology is both fascinating and impactful, with roots extending back to the early 20th century. Initially, OCR was created to assist individuals with visual impairments by converting printed words into audible formats, representing a pioneering effort to make information more accessible. The earliest OCR systems were mechanical devices that relied on fixed templates to recognize and interpret characters, but as technology progressed, so did the methods of character recognition. Modern OCR systems can now achieve remarkably high accuracy rates in recognizing and digitizing both printed and handwritten text from

diverse sources, such as scanned documents, photographs, and digital images. The widespread applications of OCR range from digitizing historical archives and automating data entry processes to facilitating text-to-speech technology for the visually impaired.

Steganalysis

Steganalysis refers to the process of detecting and decoding hidden information embedded within digital media. This concept has historical roots in ancient practices of secret communication and concealed writing. The term "steganography" itself is derived from Greek, meaning "covered writing," highlighting the art of hiding messages within ordinary texts or images to avoid detection. Steganography has evolved significantly with digital technology, allowing for complex methods of data embedding that remain undetectable to the untrained eye. With the rise of digital media, steganography techniques have become increasingly sophisticated, enabling users to embed hidden information within images, audio files, and video streams without altering the perceptible content.

2.3 Key Objectives

- Sensitive Data Detection: The foremost objective is to analyze images with precision to identify and extract sensitive information, including phone numbers, addresses, and account numbers, using advanced OCR technology. The OCR system will be equipped with algorithms that increase its accuracy in diverse scenarios, from varying text fonts to different image qualities.
- Hidden Message Detection: Another essential objective is to implement advanced steganalysis techniques to detect hidden messages or embedded information within images. This important functionality enables the identification of potential security threats, ensuring both the integrity of digital communications and the safety of its users.
- Image Upload and Capture Support: The design of the system will incorporate user-friendly features, allowing for seamless upload of images from users' devices or direct capture of images through a camera. This functionality will enhance the versatility of the system, making it accessible for users to analyze images derived from various sources.

2.4 Summary

This initiative envisions developing a powerful and effective image analysis tool that significantly enhances user privacy and security by effectively combining advanced OCR and steganalysis technologies.

Chapter 3: Literature Review

3.1 Introduction

With advancements in artificial intelligence and computer vision, image analysis tools have become essential in various fields, including cybersecurity, forensic investigations, and intellectual property protection. According to reports, the global digital image analysis market is expected to grow at an annual rate of over 15% in the coming years, highlighting the increasing demand for such tools.

Picture Detector is designed to provide an advanced image analysis service, enabling users to detect hidden messages or embedded information within images. This chapter explores existing platforms that offer similar functionalities in image analysis and steganography detection.

3.2 Existing Websites

3.2.1 StegExpose – Best for Detecting Steganography

StegExpose is an open-source tool designed to detect steganography techniques used to hide data within images. It employs statistical analysis to determine whether an image contains hidden information. While effective, the tool requires technical expertise to operate.

3.2.2 FotoForensics – Best for Image Forensic Analysis

FotoForensics is a specialized website for digital forensic image analysis. Users can upload images to receive detailed reports on potential modifications, such as Photoshop manipulations or hidden data. The platform includes tools like Error Level Analysis (ELA) to reveal invisible edits.

3.2.3 Forensically – Best for Digital Image Investigation

Forensically offers a powerful suite of tools for image analysis, including hidden data detection, digital manipulation identification, and metadata extraction. Its user-friendly interface makes it a suitable choice for both beginners and professionals.

3.2.4 DeepAI Image Recognition – Best for AI-Powered Analysis

DeepAI provides an image recognition service powered by artificial intelligence, allowing users to identify objects, text, and patterns within images. It is widely used in academic research, security applications, and digital investigations.

3.2.5 ExifTool

ExifTool is a powerful tool for reading, writing, and editing metadata in images. It can extract hidden information like GPS coordinates, camera details, and more from image files, and it can be used to analyze hidden metadata in images.

3.2.6 StegOnline

StegOnline is a web-based tool for steganography analysis. It helps detect hidden data within images, such as text or files.

3.2.7 Image Edited

A tool to detect if an image has been edited or manipulated. It can also reveal hidden details in images.

3.2.8 Google Vision API

Google's Vision API can analyze images and extract text (OCR), detect objects, and more. It can be used to extract numbers or emails from images.

3.2.9 TinEye

TinEye is a reverse image search engine that can find where an image appears online. It can also help detect manipulated or altered images.

3.2.10 Izitru

A tool for verifying images to detect any modifications or manipulations that may have been applied to the image. It uses advanced image analysis techniques to uncover hidden information within images.

Website link: <https://www.izitru.com/>

3.2.11 Amazon Rekognition

Amazon's AI service for image analysis, which allows you to detect hidden text and encrypted data within images. It provides advanced techniques for recognizing objects and elements within images.

Website link: <https://aws.amazon.com/rekognition/>

3.2.12 Google lens

A Google service that allows you to capture an image using your camera and use AI to analyze the image, extract hidden information, or help identify content within the image.

Website link: <https://lens.google/>

3.2.13 Microsoft Azure Computer Vision

Microsoft Azure Computer Vision

This site provides advanced image analysis tools via the camera using AI to detect hidden content within images and recognize text.

Website link: <https://azure.microsoft.com/en-us/services/cognitive-services/computer-vision/>

3.2.14 Tesseract OCR

An open-source OCR library for analyzing images and extracting hidden text using OCR techniques. It can be used in different applications to extract hidden text from images.

Website link: <https://github.com/tesseract-ocr/tesseract>

3.3 conclusion

In the previous sections, we presented some hidden picture analysis sites sharing the same goal of our project . We adopted some services from these applications, and we added new others to the picture detection site.

Chapter 4: Requirement Analysis

4.1 Introduction

This chapter provides a detailed analysis of the requirements for the proposed system. The system is designed to analyze images for sensitive data, detect hidden messages, and alert users before sharing potentially private information. The requirements are divided into functional and non-functional categories to ensure a comprehensive understanding of the system's capabilities and constraints

4.2 Requirement Specifications

4.2.1 Functional Requirements

The functional requirements define the core functionalities that the system must provide:

1. Image Analysis for Sensitive Data
 - The system should be able to extract and analyze text from images using Optical Character Recognition (OCR).
 - It should detect sensitive information, such as:

- Phone numbers
 - Addresses
 - Bank account numbers
 - ID numbers
2. Detection of Hidden Messages in Images
 - The system should use Steganalysis techniques to identify concealed messages or embedded data within images.
 - It should support different steganography detection methods, such as:
 - Pixel analysis
 - Metadata examination
 - Noise pattern recognition
 3. Image Upload and Capture Support
 - Users should be able to upload an image from their device for analysis.
 - The system should allow users to capture an image using the device's camera for real-time analysis.
 4. User Alert and Data Masking
 - If sensitive data or hidden messages are detected, the system should display a warning message to the user.
 - Users should have the option to automatically blur, redact, or remove sensitive data before sharing the image.

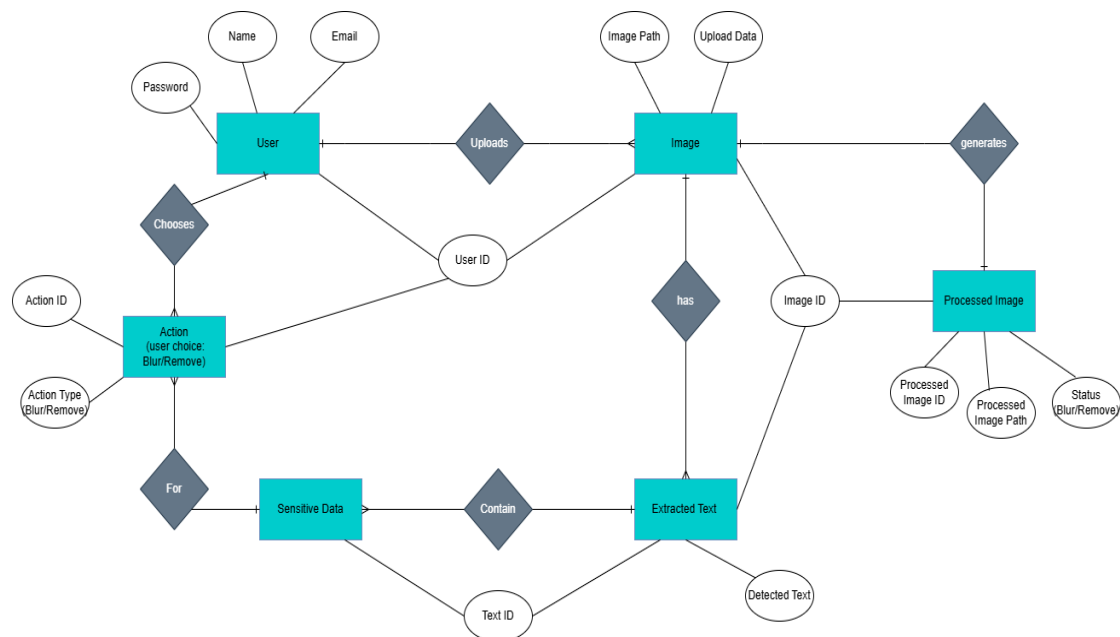
4.2.2 Non-functional Requirements

The non-functional requirements define the system's performance, security, and usability constraints:

1. Performance
 - The system should analyze an image and return results within 5 seconds for standard-sized images (e.g., less than 5 MB).
 - The OCR and Steganalysis modules should be optimized for real-time processing.
2. Security
 - All uploaded images should be processed locally or stored temporarily with automatic deletion after processing.
 - The system must encrypt any stored or transmitted data to ensure user privacy.
3. Usability

- The interface should be simple and user-friendly, with clear options for uploading, analyzing, and modifying images.
 - Users should receive intuitive alerts and recommendations based on detected data.
4. Scalability
 - The system should be able to handle multiple image requests concurrently without performance degradation.
 5. Platform Compatibility
 - The system should support multiple platforms, including web browsers, mobile apps (iOS & Android), and desktop applications.

4.3 Entity Relationship Diagram (ERD):



4.3.1 General ERD:

An Entity Relationship Diagram (ERD) is a visual representation of the relationships between different entities in a database. It helps in designing and structuring a database by showing how entities (such as people, products, or orders) relate to each other.

Key Components of an ERD:

1. Entities – Represent real-world objects or concepts (e.g., "Customer," "Order").
2. Attributes – Describe properties of an entity (e.g., a "Customer" entity may have "Name," "Email").
3. Relationships – Show how entities are connected (e.g., "Customer places Order").
4. Primary Keys (PK) – Uniquely identify each entity instance.
5. Foreign Keys (FK) – Reference a primary key from another entity to establish relationships.

Types of Relationships in ERD:

- One-to-One (1:1) – One entity instance is related to only one instance of another entity.
- One-to-Many (1:M) – One entity instance is related to multiple instances of another entity.
- Many-to-Many (M:N) – Multiple instances of one entity are related to multiple instances of another.

4.4 Conclusion:

The development of this website marks a significant advancement in digital privacy protection. By integrating OCR, pattern recognition, and image processing, it provides a robust solution for detecting and sanitizing sensitive information in images before sharing. This proactive approach empowers users to control their digital footprint and mitigate risks such as identity theft and financial fraud. With its scalable and user-friendly design, the website addresses both individual and organizational needs, reinforcing the importance of secure digital communication. As data privacy concerns continue to grow, this project serves as a vital tool in safeguarding sensitive information in an increasingly interconnected world.